

Problem Set 1 - Suggested Solutions

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0.1 Sum Stats and Institutional Details

0.1.1 Do some brief diligence on the products and industry. What do you anticipate may be some important determinants of demand, substitution, and pricing?

Answers may vary. Students may include information on conditions treated by the medications, the prevalence of these conditions, or relevant (perceived or otherwise) differences between the products. Responses may also include general information on popularity or use trends for the products.

0.1.2 Complete the table above by adding columns for the mean: market share, unit price, price/100 tablets, and unit wholesale price. Interpret any notable patterns you see in the summary statistics.

```
[1]: import numpy as np
import pandas as pd
from linearmodels.iv import IV2SLS
import statsmodels.api as sm
import statsmodels.formula.api as smf
otc = pd.read_csv('OTC_Sales.csv')
```

```
[2]: # calculate total sales for each store-week pair
product_numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
otc['total_sales'] = otc[[f'sales_{number}' for number in product_numbers]].
    ↪sum(axis = 1)
```

```
[3]: # reshape to make it easier to work with
otc = pd.wide_to_long(otc, ['sales', 'price', 'cost', 'prom'],
                      i = ['store', 'week'], j = 'product', sep = '_').
    ↪reset_index()
summary = otc.groupby('product').agg(
    price = ('price', 'mean'),
    total_sales = ('sales', 'sum'),
    wholesale_price = ('cost', 'mean')
).reset_index()

# add given information so that table is easy to read
brand_names = {
    1: 'Tylenol',
```

```

2: 'Tylenol',
3: 'Tylenol',
4: 'Advil',
5: 'Advil',
6: 'Advil',
7: 'Bayer',
8: 'Bayer',
9: 'Bayer',
10: 'Store Brand',
11: 'Store Brand'}
size = {
  1: 25,
  2: 50,
  3: 100,
  4: 25,
  5: 50,
  6: 100,
  7: 25,
  8: 50,
  9: 100,
  10: 50,
  11: 100
}
summary['brand'] = summary['product'].map(brand_names)
summary['size'] = summary['product'].map(size)

# calculate new variables
summary['market_share'] = summary['total_sales'] / (summary['total_sales'].
↪sum())
summary['price_per_100'] = (summary['price'] / summary['size']) * 100

# reorder columns and round for clarity
table_1 = summary[['product', 'brand', 'size', 'market_share',
                    'price', 'price_per_100', 'wholesale_price']].round(2)
table_1

```

```

[3]:
   product    brand  size  market_share  price  price_per_100  \
0         1  Tylenol   25           0.14   3.43           13.71
1         2  Tylenol   50           0.18   4.95           9.89
2         3  Tylenol  100           0.12   7.03           7.03
3         4   Advil   25           0.12   2.97          11.88
4         5   Advil   50           0.08   5.15          10.29
5         6   Advil  100           0.04   8.16           8.16
6         7   Bayer   25           0.04   2.67          10.70
7         8   Bayer   50           0.03   3.62           7.24
8         9   Bayer  100           0.08   3.97           3.97
9        10 Store Brand   50           0.09   1.94           3.87

```

10	11	Store Brand	100	0.08	4.43	4.43
		wholesale_price				
0		2.19				
1		3.68				
2		5.77				
3		2.03				
4		3.63				
5		6.10				
6		1.85				
7		2.44				
8		3.71				
9		0.91				
10		1.91				

0.2 Logit Demand Estimation

Consider the utility function for product j in store-week t for consumer i :

$$u_{ijt} = -p_{jt} + X_{jt} + \xi_{jt} + \epsilon_{ijt} \quad (1)$$

where p_{jt} is price, X_{jt} are other observed product characteristics, ξ_{jt} are unobserved product characteristics, and ϵ_{ijt} is an i.i.d. EV1 logit consumer-product-market unobservable.

Estimate this model:

0.2.1 Using OLS with price, promotion, and an indicator for whether the product is a “store brand” as product characteristics.

```
[4]: # need shares for each observation
otc['share'] = otc['sales'] / otc['count']
otc['outside_share'] = 1 - (otc['total_sales'] / otc['count'])

# indicator for store brand
otc['store_brand'] = np.where(otc['product'] > 9, 1, 0)
otc['size'] = otc['product'].map(size)
```

```
[5]: # create dependent variable: ln(share - outside_share)
otc['ln_share'] = np.log(otc['share']) - np.log(otc['outside_share'])
```

```
[6]: model = smf.ols('ln_share ~ price + prom + store_brand', data = otc)
logit_modela = model.fit()
logit_modela.summary()
```

[6]:

Dep. Variable:	ln_share	R-squared:	0.026
Model:	OLS	Adj. R-squared:	0.026
Method:	Least Squares	F-statistic:	335.2
Date:	Wed, 19 Feb 2025	Prob (F-statistic):	8.60e-215
Time:	13:31:23	Log-Likelihood:	-48633.
No. Observations:	37290	AIC:	9.727e+04
Df Residuals:	37286	BIC:	9.731e+04
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-7.6453	0.014	-552.117	0.000	-7.672	-7.618
price	-0.0662	0.003	-24.543	0.000	-0.071	-0.061
prom	0.1990	0.016	12.316	0.000	0.167	0.231
store_brand	-0.2672	0.013	-21.180	0.000	-0.292	-0.242

Omnibus:	2157.310	Durbin-Watson:	1.362
Prob(Omnibus):	0.000	Jarque-Bera (JB):	2189.616
Skew:	-0.552	Prob(JB):	0.00
Kurtosis:	2.563	Cond. No.	18.2

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

0.2.2 Using OLS with price and promotion as product characteristics and product fixed effects (where a “product” is a brand-size combination).

```
[7]: model = smf.ols('ln_share ~ price + prom + C(product)', data = otc)
logit_modelb = model.fit()
logit_modelb.summary()
```

[7]:

Dep. Variable:	ln_share	R-squared:	0.457
Model:	OLS	Adj. R-squared:	0.457
Method:	Least Squares	F-statistic:	2614.
Date:	Wed, 19 Feb 2025	Prob (F-statistic):	0.00
Time:	13:31:23	Log-Likelihood:	-37744.
No. Observations:	37290	AIC:	7.551e+04
Df Residuals:	37277	BIC:	7.562e+04
Df Model:	12		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-6.0693	0.037	-164.587	0.000	-6.142	-5.997
C(product)[T.2]	0.6879	0.023	30.566	0.000	0.644	0.732
C(product)[T.3]	0.9188	0.040	22.778	0.000	0.840	0.998
C(product)[T.4]	-0.4312	0.017	-25.661	0.000	-0.464	-0.398
C(product)[T.5]	-0.1960	0.024	-8.178	0.000	-0.243	-0.149
C(product)[T.6]	0.0045	0.051	0.088	0.930	-0.096	0.105
C(product)[T.7]	-1.6597	0.018	-93.199	0.000	-1.695	-1.625
C(product)[T.8]	-1.5844	0.016	-96.398	0.000	-1.617	-1.552
C(product)[T.9]	-0.5523	0.018	-31.542	0.000	-0.587	-0.518
C(product)[T.10]	-1.2360	0.022	-56.168	0.000	-1.279	-1.193
C(product)[T.11]	-0.7387	0.019	-38.296	0.000	-0.777	-0.701
price	-0.3404	0.010	-33.375	0.000	-0.360	-0.320
prom	0.3220	0.013	25.508	0.000	0.297	0.347
Omnibus:	2680.344		Durbin-Watson:	1.901		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	3703.376		
Skew:	-0.621		Prob(JB):	0.00		
Kurtosis:	3.916		Cond. No.	112.		

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

0.2.3 Estimate the models of (a) and (b) using the Hausman instrument (average price in other markets).

```
[8]: def get_instruments(df):
    # compute average prices per store, week, and product
    avg_prices = df.groupby(['store', 'week', 'product'])['price'].mean().
    ↪reset_index()

    # merge to bring average prices into the original dataframe
    df = df.merge(avg_prices, on=['store', 'week', 'product'], suffixes=('_',
    ↪'_avg'))

    # compute instrument: average price across stores excluding the current
    ↪store
    df['instrument'] = df.groupby(['week', 'product'])['price_avg'].
    ↪transform(lambda x: (x.sum() - x) / (x.count() - 1))

    return df.drop(columns=['price_avg'])

otc = get_instruments(otc)
```

```
[9]: # model a using IV
formula = ('ln_share ~ 1 + prom + store_brand + [price ~ instrument]')
logit_modela_iv = IV2SLS.from_formula(formula, otc).fit()
logit_modela_iv.summary
```

[9]:

Dep. Variable:	ln_share	R-squared:	0.0263
Estimator:	IV-2SLS	Adj. R-squared:	0.0262
No. Observations:	37290	F-statistic:	874.64
Date:	Wed, Feb 19 2025	P-value (F-stat)	0.0000
Time:	13:31:23	Distribution:	chi2(3)
Cov. Estimator:	robust		

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
Intercept	-7.6453	0.0143	-533.36	0.0000	-7.6734	-7.6172
prom	0.1990	0.0158	12.585	0.0000	0.1680	0.2300
store_brand	-0.2672	0.0136	-19.635	0.0000	-0.2938	-0.2405
price	-0.0662	0.0028	-23.364	0.0000	-0.0718	-0.0607

Endogenous: price

Instruments: instrument

Robust Covariance (Heteroskedastic)

Debiased: False

```
[10]: # model b using IV
otc_p = pd.get_dummies(otc, columns = ['product'])
fixed_effects = " + ".join([col for col in otc_p.columns if col.
    ↳startswith("product_")])

formula = f'ln_share ~ prom + {fixed_effects} + [price ~ instrument]'
logit_modelb_iv = IV2SLS.from_formula(formula, otc_p).fit()
logit_modelb_iv.summary
```

[10]:

Dep. Variable:	ln_share	R-squared:	0.4520
Estimator:	IV-2SLS	Adj. R-squared:	0.4519
No. Observations:	37290	F-statistic:	6.581e+06
Date:	Wed, Feb 19 2025	P-value (F-stat)	0.0000
Time:	13:31:23	Distribution:	chi2(13)
Cov. Estimator:	robust		

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
	prom	0.2648	18.676	0.0000	0.2370	0.2926
	product_1	-5.4225	0.0531	-102.16	-5.5265	-5.3184
	product_2	-4.4459	0.0763	-58.246	-4.5955	-4.2963
	product_3	-3.8220	0.1081	-35.352	-4.0338	-3.6101
	product_4	-5.9368	0.0468	-126.82	-6.0286	-5.8451
	product_5	-5.2923	0.0797	-66.391	-5.4485	-5.1361
	product_6	-4.5235	0.1256	-36.002	-4.7698	-4.2773
	product_7	-7.2170	0.0432	-167.18	-7.3016	-7.1324
	product_8	-6.9629	0.0566	-122.93	-7.0739	-6.8518
	product_9	-5.8604	0.0629	-93.166	-5.9837	-5.7371
	product_10	-6.9324	0.0337	-205.62	-6.9985	-6.8664
	product_11	-5.9682	0.0683	-87.364	-6.1021	-5.8343
	price	-0.5286	0.0153	-34.607	-0.5586	-0.4987

Endogenous: price

Instruments: instrument

Robust Covariance (Heteroskedastic)

Debiased: False

0.2.4 Compute the mean own-price elasticities for all products

```
[11]: alpha = logit_modelb_iv.params.price
otc['elasticity'] = alpha * (otc['price']) * (1 - otc['share'])

# add this to our summary table so that we can easily compare
sum_elast = otc.groupby('product').agg(elasticity = ('elasticity', 'mean')).
    ↪reset_index()
summary = pd.merge(summary, sum_elast, how = 'outer')
summary.round(3)
```

```
[11]:
```

	product	price	total_sales	wholesale_price	brand	size	\
0	1	3.427	51280	2.187	Tylenol	25	
1	2	4.946	63604	3.681	Tylenol	50	
2	3	7.028	41305	5.771	Tylenol	100	
3	4	2.969	42032	2.027	Advil	25	
4	5	5.146	27627	3.630	Advil	50	
5	6	8.158	12703	6.104	Advil	100	
6	7	2.674	14365	1.851	Bayer	25	
7	8	3.619	12310	2.438	Bayer	50	
8	9	3.971	28324	3.710	Bayer	100	
9	10	1.935	33175	0.910	Store Brand	50	
10	11	4.427	27208	1.914	Store Brand	100	

	market_share	price_per_100	elasticity
0	0.145	13.708	-1.810
1	0.180	9.893	-2.612

2	0.117	7.028	-3.713
3	0.119	11.876	-1.568
4	0.078	10.291	-2.719
5	0.036	8.158	-4.312
6	0.041	10.697	-1.413
7	0.035	7.238	-1.913
8	0.080	3.971	-2.098
9	0.094	3.871	-1.023
10	0.077	4.427	-2.339

0.3 New Product Introduction

The chain of stores you have data from is considering introducing a 25 tablet size bottle.

0.3.1 Assume WTP for the new product will be equal to average WTP of the brand name 25 tablet products, minus the “store brand” effect you estimated in the first part above. Assume there are no promotions of the new product. What will be the expected demand for the new product at a price of \$2.00?

```
[12]: # WTP = average product_fe - store_brand
beta_product = {}
for i in product_numbers:
    beta_product[i] = logit_modelb_iv.params.get(f"product_{i}", 1)
beta_fe = pd.Series(beta_product)

beta_storebrand = logit_modela_iv.params.store_brand

WTP = ((beta_product[1] + beta_product[4] + beta_product[7])/3) -
    beta_storebrand
```

```
[13]: # pull coefficients for use in calculating shares
beta_prom = logit_modelb_iv.params.prom
alpha = logit_modelb_iv.params.price

beta_product = {}
for i in product_numbers:
    beta_product[i] = logit_modelb_iv.params.get(f"product_{i}", 1)

beta_fe = pd.Series(beta_product)
```

```
[14]: # recalculate market shares with new good
# calculate logit share numerator for each good
numerator = {}
for i in product_numbers:
    otc_i = otc[otc['product'] == i]
    numerator[i] = np.exp((alpha * otc_i['price'].mean()) + (beta_prom *
        otc_i['prom'].mean()) +
```



```

        (beta_storebrand * otc_i['store_brand'].mean()) +
        beta_fe[i])
numerator = pd.Series(numerator)

# use WTP calculated above as product FE
numerator[12] = np.exp((alpha * 2) + (beta_prom * 0) + (beta_storebrand * 1) +
        WTP)

denominator = 1 + numerator.sum()

```

```

[15]: # calculate total customers across all stores in all weeks
total_traffic = (otc.groupby(['store', 'week'])['count'].first()).sum()

# estimate sales of each good before and after new product introduction
new_market_shares = numerator / denominator
sales_new = new_market_shares * total_traffic
old_market_shares = otc.groupby('product')['share'].mean()

#add in value of 0 for new product before it was introduced
old_market_shares[12] = 0
sales_old = old_market_shares * total_traffic

```

```

[16]: # put it all into a table
otc_new = pd.DataFrame({
    'Old Shares': old_market_shares.values,
    'Old Sales': sales_old.values.round(2),
    'New Shares': new_market_shares.values,
    'New Sales': sales_new.values.round(2)},
    index = new_market_shares.index
)

# Rename the index to 'Products'
otc_new.index.name = 'Products'
otc_new

```

```

[16]:

```

	Old Shares	Old Sales	New Shares	New Sales
Products				
1	0.000771	51361.08	0.000724	48249.15
2	0.000947	63119.78	0.000872	58098.00
3	0.000621	41386.19	0.000545	36291.64
4	0.000642	42747.23	0.000559	37273.48
5	0.000424	28240.52	0.000337	22431.84
6	0.000197	13110.40	0.000149	9900.88
7	0.000218	14557.37	0.000185	12325.76
8	0.000184	12246.41	0.000145	9688.80
9	0.000430	28681.10	0.000371	24688.23
10	0.000495	33000.98	0.000278	18518.13

11	0.000408	27204.97	0.000194	12906.47
12	0.000000	0.00	0.000707	47109.12

0.3.2 Break down the benefits and costs to the store of introducing this new format (assume wholesale price is \$1.00 and there are no fixed or other variable costs of the new product introduction).

```
[17]: # pull price and cost information and add in the new product
price = otc.groupby('product')['price'].mean()
cost = otc.groupby('product')['cost'].mean()
price[12] = 2
cost[12] = 1

# add to otc_new
otc_new['Price'] = price.values.round(2)
otc_new['Cost'] = cost.values.round(2)

[18]: # calculate profit before and after the new product
otc_new['Old Profit'] = ((otc_new['Price'] - otc_new['Cost']) * (otc_new['Old_
↪Sales'])).round(2)
otc_new['New Profit'] = ((otc_new['Price'] - otc_new['Cost']) * (otc_new['New_
↪Sales'])).round(2)

print(f'Profit before the new product: {otc_new['Old Profit'].sum().round(2)}')
print(f'Profit after the new product: {otc_new['New Profit'].sum().round(2)}')
print(f'Therefore, introducing the new product changes profit by_
↪{((otc_new['New Profit'].sum()) - (otc_new['Old Profit'].sum())).round(2)}')
```

Profit before the new product: 442504.29

Profit after the new product: 395536.09

Therefore, introducing the new product changes profit by -46968.2

0.4 Information Intervention

The chain of stores you have data from is considering a campaign to help educate customers that there is no efficacy difference between the brand name and “store brand” drugs.

0.4.1 Assume the campaign increases the WTP for the “store brands” by the full absolute value of the “store brand” effect you estimated in the first part. What is the net benefit and cost to the store? To consumers?

```
[19]: # pull fe again (just in case)
beta_product = {}
for i in product_numbers:
    beta_product[i] = logit_modelb_iv.params.get(f"product_{i}", 1)
beta_fe = pd.Series(beta_product)
```

```
beta_storebrand = logit_modela_iv.params.store_brand
```

```
[20]: # calculate CS before change
# calculate logit share numerator for each good - increase by store brand since
# consumers received more utility than they realized
numerator = {}
for i in product_numbers:
    otc_i = otc[otc['product'] == i]
    numerator[i] = np.exp((alpha * otc_i['price'].mean()) + (beta_prom *
# otc_i['prom'].mean())
                                + beta_fe[i] + (np.abs(beta_storebrand) *
# otc_i['store_brand'].mean()))

numerator_before = pd.Series(numerator)
denominator_before = 1 + numerator_before.sum()
```

```
[21]: # recalculate WTP without the store-brand effect
beta_fe[10] = beta_fe[10] + np.abs(beta_storebrand)
beta_fe[11] = beta_fe[11] + np.abs(beta_storebrand)
```

```
[22]: # recalculate market shares with new WTP
# calculate logit share numerator for each good
numerator = {}
for i in product_numbers:
    otc_i = otc[otc['product'] == i]
    numerator[i] = np.exp((alpha * otc_i['price'].mean()) + (beta_prom *
# otc_i['prom'].mean())
                                + beta_fe[i] + (np.abs(beta_storebrand) *
# otc_i['store_brand'].mean()))

numerator_after = pd.Series(numerator)
denominator_after = 1 + numerator_after.sum()
```

```
[23]: # estimate sales of each good before and after information campaign
new_market_shares = numerator_after / denominator_after
sales_new = new_market_shares * total_traffic
old_market_shares = otc.groupby('product')['share'].mean()
sales_old = old_market_shares * total_traffic
```

```
[24]: # put it all into a table
otc_info = pd.DataFrame({
    'Old Shares': old_market_shares.values,
    'Old Sales': sales_old.values.round(2),
    'New Shares': new_market_shares.values,
    'New Sales': sales_new.values.round(2)},
    index = new_market_shares.index
)
```

```
# Rename the index to 'Products'
otc_info.index.name = 'Products'
otc_info
```

```
[24]:
```

	Old Shares	Old Sales	New Shares	New Sales
Products				
1	0.000771	51361.08	0.000724	48255.30
2	0.000947	63119.78	0.000872	58105.40
3	0.000621	41386.19	0.000545	36296.27
4	0.000642	42747.23	0.000559	37278.23
5	0.000424	28240.52	0.000337	22434.70
6	0.000197	13110.40	0.000149	9902.15
7	0.000218	14557.37	0.000185	12327.33
8	0.000184	12246.41	0.000145	9690.03
9	0.000430	28681.10	0.000371	24691.37
10	0.000495	33000.98	0.000620	41280.37
11	0.000408	27204.97	0.000432	28770.93

```
[25]: # pull price and cost information and add in the new product
price = otc.groupby('product')['price'].mean()
cost = otc.groupby('product')['cost'].mean()

# add to otc_info
otc_info['Price'] = price.values.round(2)
otc_info['Cost'] = cost.values.round(2)
```

```
[26]: # calculate profit before and after the new product
otc_info['Old Profit'] = ((otc_info['Price'] - otc_info['Cost']) *
    ↳ (otc_info['Old Sales'])).round(2)
otc_info['New Profit'] = ((otc_info['Price'] - otc_info['Cost']) *
    ↳ (otc_info['New Sales'])).round(2)

print(f'Profit before the new product: {otc_info['Old Profit'].sum().round(2)}')
print(f'Profit after the new product: {otc_info['New Profit'].sum().round(2)}')
print(f'Therefore, introducing the new product changes profit by
    ↳ {(otc_info['New Profit'].sum()) - (otc_info['Old Profit'].sum())}.
    ↳ round(2)}')
```

Profit before the new product: 442504.29

Profit after the new product: 411888.37

Therefore, introducing the new product changes profit by -30615.92

```
[27]: # CS = log(denominator)/alpha
CS_before = np.log(denominator_before) / alpha
print(f'Consumer Surplus before the information campaign is {CS_before}')

CS_after = np.log(denominator_after) / alpha
```

```
print(f'Consumer Surplus after the information campaign is {CS_after}')
```

Consumer Surplus before the information campaign is -0.008898765290945184

Consumer Surplus after the information campaign is -0.009365139219042605